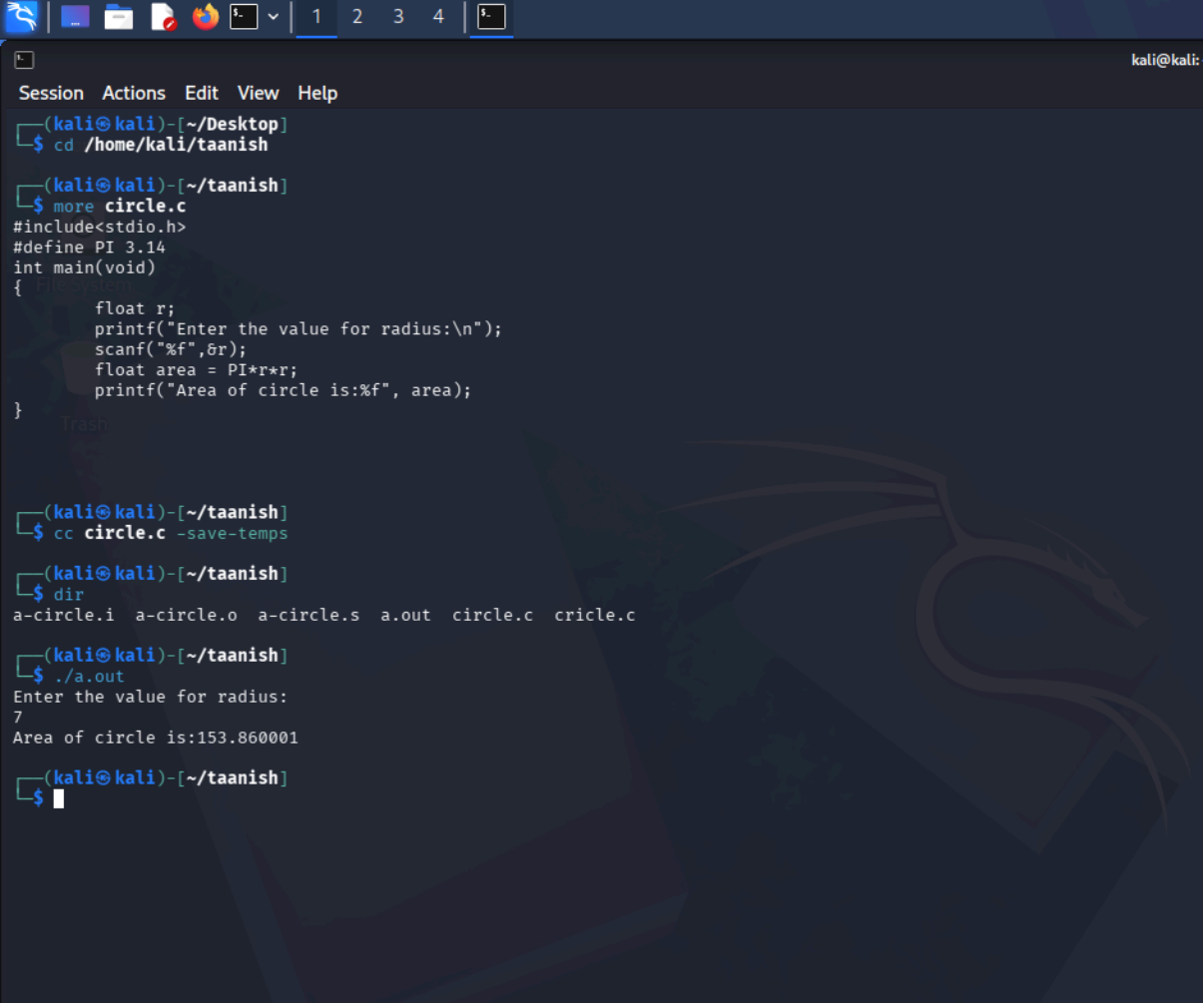


Experiment-2

Q. Finding the area of a circle after getting input for the radius from the user.

A.



```
(kali@kali)-[~/Desktop]
$ cd /home/kali/taanish

(kali@kali)-[~/taanish]
$ more circle.c
#include<stdio.h>
#define PI 3.14
int main(void)
{
    float r;
    printf("Enter the value for radius:\n");
    scanf("%f",&r);
    float area = PI*r*r;
    printf("Area of circle is:%f", area);
}

(kali@kali)-[~/taanish]
$ cc circle.c -save-temps

(kali@kali)-[~/taanish]
$ dir
a-circle.i a-circle.o a-circle.s a.out circle.c cricle.c

(kali@kali)-[~/taanish]
$ ./a.out
Enter the value for radius:
7
Area of circle is:153.860001

(kali@kali)-[~/taanish]
$
```

By passing specific flags to the compiler (like `-save-temps`), you can force it to save the temporary files generated at each step.

1. `.c` (Source Code File) -This is the original, human-readable source code written in the C programming language. It is the starting point of the compilation process.
2. `.i` (pre-processed File)- This is the output of the Preprocessor. Before translating the code, the compiler reads the `.c` file and resolves all preprocessor directives (lines starting with `#`). It expands macros (like `#define`), includes the contents of header files (like `#include`

`<stdio.h>`), and strips out all the comments. The resulting `.i` file is still C code, but it is usually much larger than the original `.c` file.

3. `.s` (Assembly file)- This is the output of the Compiler in the strict sense. The compiler takes the preprocessed `.i` file and translates the C code into architecture-specific Assembly language. Assembly is a low-level programming language that is human-readable but directly corresponds to the processor's machine instructions.
4. `.o` (Object File)-This is the output of the Assembler. The assembler takes the `.s` assembly file and converts it into machine code (binary code containing 1s and 0s) that the CPU can understand directly. While it contains executable machine instructions, it cannot be run on its own yet because it hasn't been linked to standard libraries (like the C standard library for `printf`).
5. `.out` (Linked Executable file)- This is the final output produced by the Linker. The linker takes one or more object (`.o`) files and links them together with any necessary external software libraries. a `.out` (which stands for "assembler output") is the default name given to the final executable binary file on Unix-like operating systems if you don't explicitly specify an output name.

To explain the above thing in short, we can say that,
Source (`.c`) → Preprocessed (`.i`) → Assembly (`.s`) → Object (`.o`) → Executable (`a.out`).

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